



SDE Intern Assignment

Personalized News, Media & Social Aggregator

A high-performance Next.js 14 feed engine combining real-time technology news, TMDB entertainment discovery, and Mastodon social threads into a single unified workspace.



LIVE DEMO
mypulse.vercel.app



GITHUB REPOSITORY
github.com/ChethanNazre/Pulse



TECH STACK
Next.js 14 • React 18 • Redux Toolkit

Core Engineering Objectives



Unified Multi-Stream

Consolidates real-time news, movie discovery, and Mastodon social feeds into a single unified, coherent dashboard.



Dynamic Curation

Empowers users with 400ms debounced search, topic category pills, drag-and-drop card reordering, and offline bookmarking.

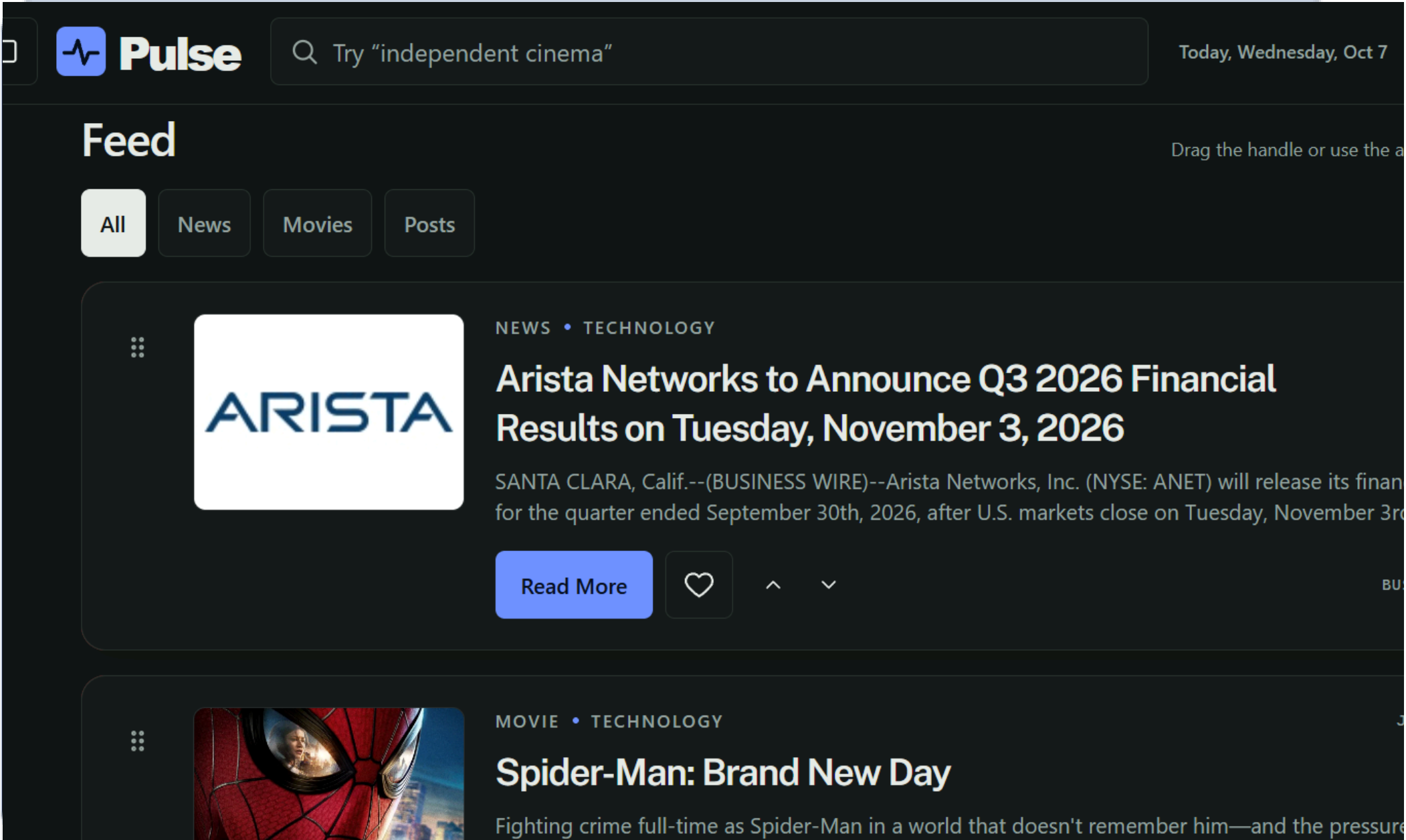


Resilient UX Boundaries

Eliminates dead ends through server-side key security, per-source error boundaries, loading skeletons, and responsive layouts.

Desktop Discovery Workspace

Dark Theme Interface



Smart Search & Topic Filters

Instant topic switching across **All**, **News**, **Movies**, and **Posts** with live query filtering.

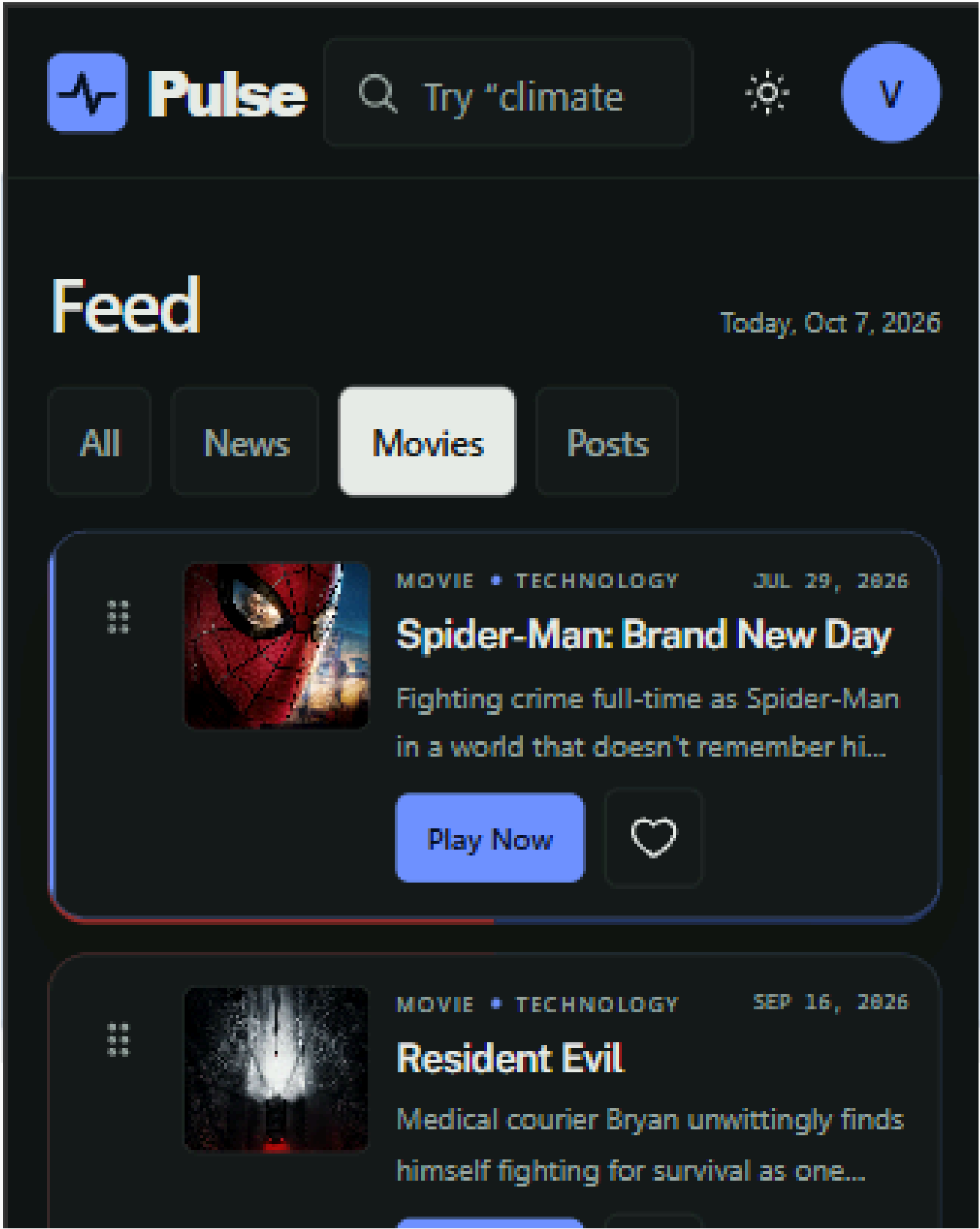
Interactive Card Reordering

Drag handle controls and vertical arrows allow instant feed prioritization on the fly.

One-Click Favorites & CTAs

Direct links to full news sources, movie trailers, and local offline bookmarking.

Mobile Feed & Touch Interface



Compact Bottom Navigation Bar

Dedicated touch targets for **Feed**, **Trending**, **Favorites**, and **Settings** for single-handed usage.



Actionable Media Cards

Optimized thumbnail ratios with direct **Play Now** actions and quick bookmark toggles.



Fluid Touch Layout

Smooth vertical scrolling, zero horizontal overflow, and auto-hiding search headers.

Integrated Multi-API Ecosystem



Technology & News

- ✓ **Server Proxy:** /api/news route
- ✓ **Topic Allowlist:** High-credibility tech feeds
- ✓ **Rich Metadata:** Publisher & publishing date



TMDb Movies & Shows

- ✓ **Media Endpoint:** /api/movies route
- ✓ **Trending & Search:** Posters & overviews
- ✓ **Interaction:** Direct movie streaming links



Decentralized Social

- ✓ **Social Stream:** /api/social route
- ✓ **Public Timelines:** Mastodon tag feeds
- ✓ **Sanitization:** HTML stripping & clean text

Search Engine & Topic Controls



400 ms Debounced Search

Eliminates rapid API call spamming while typing queries like *"independent cinema"* or *"climate"*.



6 Core Topic Categories

Instant filtering across Technology, Sports, Finance, Entertainment, Health, and Science.



Source Type Pill Filters

Filter the main stream dynamically down to **All**, **News**, **Movies**, or **Posts** in real-time.



Zero-Reload Redux State

Instant UI category updates executed entirely client-side without full page refreshes.

Feed Personalization & Card Reordering



Framer Motion Drag

Smooth vertical drag handles on every feed card for customizable reading order.



Precision Arrow Nudge

Up and Down buttons provide explicit keyboard and accessible reorder alternatives.



"See Fewer" Feedback

Allows users to deemphasize specific feed items and learn preference weighting.



Layout Storage Sync

Customized card arrangements persist across browser reloads automatically.

LocalStorage & Offline Favorites

LocalStorage Vault

★ **Favorites Vault:** Bookmarked articles and movies saved offline.

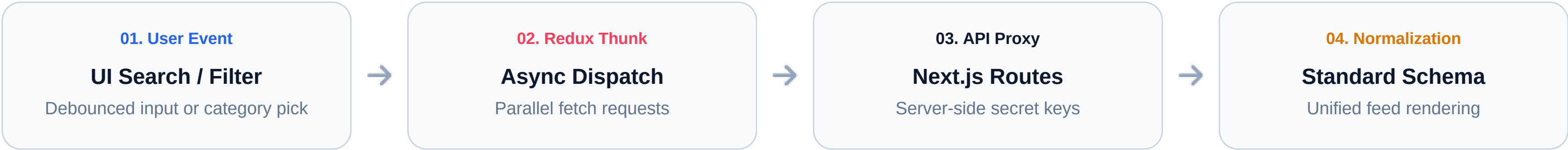
🌙 **Theme State:** Dark/Light theme mode selection persistence.

⚙️ **Card Preferences:** Custom hidden sources and item positions.

Offline Accessibility

Saved items remain 100% accessible on the **Favorites** view even when network connectivity drops or when APIs undergo rate-limit resets.

End-to-End Data Pipeline Flow



Server-Side Secret Key Protection



API Key Isolation

NEWS_API_KEY and TMDB_API_KEY remain strictly on the Next.js server layer and are never exposed to browser bundles.



CORS & Bounded Queries

Internal same-origin proxy handlers prevent malicious external requests and limit query parameters to pre-approved categories.



Input Sanitization

Strips HTML tags from Mastodon social feeds and validates JSON schemas before dispatching items to Redux state.

Zero Dead-End Error Boundaries



Skeleton Loaders

Animated skeleton cards prevent layout shifts and maintain user context while asynchronous API requests resolve.



Isolated API Faults

If TMDB or News API experiences an outage, remaining healthy feeds continue serving content without crashing the app.



Informative Empty States

Clear, helpful user guidance screens display when search queries produce zero matches or when favorites list is empty.

Performance & Inclusive Design



IntersectionObserver Feed

Smooth auto-loading continuous scrolling with a manual "Load More" fallback trigger.



Keyboard Navigation (a11y)

Full focus-ring visibility, ARIA accessibility attributes, and keyboard card controls.



Reduced Motion Fallbacks

Respects OS-level prefers-reduced-motion settings by replacing complex CSS animations.



System Theme Sync

Detects user OS dark mode preference on initial page load and persists user choices.

Testing Suite & Deployment Commands

Test Suites Included

Jest & React Testing Library

Unit and integration suites testing Redux slices and component rendering.

Playwright E2E Setup

End-to-end user journey automation tests configured in codebase repository.

> Terminal Deployment

Node 20+

1. Clone repository

```
git clone https://github.com/ChethanNazre/Pulse.git
```

2. Install dependencies & start server

```
npm install && npm run dev
```

3. Execute test suite

```
npm run test
```


Next Phase Enhancements



SSE Real-Time Refresh

Implementation of Server-Sent Events to push 30-second background feed updates without full polling.



Supabase Multi-Device Auth

Transitioning local preferences to cloud authentication for seamless bookmark sync across mobile and web.



i18n Multi-Language

Native internationalization switcher support (EN / ES / DE) for localized feeds and interface elements.



PULSE

Thank You!

Questions & Architectural Discussion



mypulse.vercel.app



[GitHub Repository](#)